

Data Structure

Level 1

Singly Linked List Data Structures Operations

The background of the slide features a complex network of thin, dark lines crisscrossing over a lighter, textured surface, resembling a web or a data visualization. In the lower portion of the image, the silhouettes of several people are visible, appearing to be in a collaborative setting, possibly a meeting or a workshop, with some individuals leaning forward as if working on a large screen or table.

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Level 1

Find



Definition

Find() Function:

Searches for a node containing a specific value in the linked list.

■ Behavior:

- Starts from the head
- Compares each node's value
- Returns pointer to the node if found
- Returns NULL if not found



Explanation

- ◆ The function traverses the linked list node by node.
- ◆ If `head->value == Value`, it immediately returns the current node.
- ◆ If the loop finishes without finding the value, it returns NULL.
- ◆ This is a linear search operation.

Procedures (Insert at Beginning)

- 1 Create an empty linked list
- 2 Insert nodes using InsertAtBeginning()
- 3 Call Find(head, Value)
- 4 Traverse the list inside the function
- 5 Compare each node's value
- 6 Return node pointer if found
- 7 Return NULL if not found
- 8 Check returned pointer in main()
- 9 Print result message

Example

Inserted values (in order):
1, 2, 3, 4, 5

Actual list becomes:
 $5 \rightarrow 4 \rightarrow 3 \rightarrow 2 \rightarrow 1 \rightarrow \text{NULL}$

Searching for value 2

 Result: Node Found :-)

Note

The function returns a pointer to the node, not just a boolean value.



Comparison

■ Linked List Search

- ✓ Sequential access only
- ✓ Time Complexity: $O(n)$
- ✓ No direct indexing

■ Array Search

- ✓ Can access elements by index
- ✓ Still $O(n)$ for linear search
- ✓ Faster random access

Warning

If you forget to check NULL before accessing the returned pointer, the program may crash.

Summary

- ✓ Find() performs a linear search in the linked list
- ✓ Returns pointer to node if found
- ✓ Returns NULL if not found
- ✓ Time Complexity is $O(n)$
- ✓ Requires traversal from head to end

Thank you 😊